

# HANDWRITING TO TEXT OCR

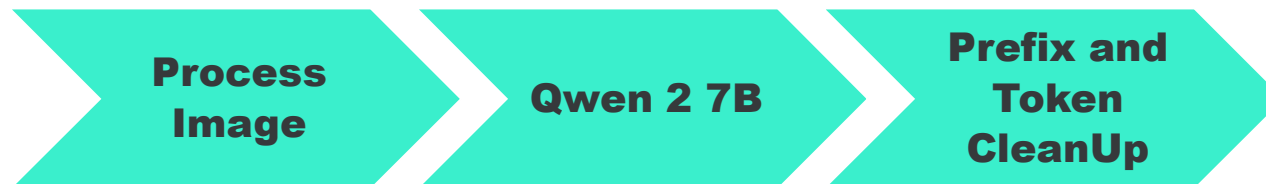


# HANDWRITING TO TEXT OCR

## MODEL 1(PURE OCR):



## MODEL 2(VL MODEL):



# OUTPUT COMPARISON

## Abstract

Research suggests that note taking is most effective when done by hand. Yes, handwritten notes. Certainly, typewritten or typed out notes are more efficient, however notes written by hand are retained by the human brain much longer.

Integrating handwritten notes into a computer based productivity workflow is tricky. Also, in terms of hardware, if all you have is a smartphone, then we need to deal with the situation a bit differently.

This talk will introduce to you a simple system to integrate handwritten notes into your org-mode (Emacs) based productivity workflow.

- Gist of the problem of dealing with handwritten notes

- Advantages of handwritten notes

- Emacs org mode workflow

- Option 1: Hardware for handwriting NOT available

- Option 2: Hardware for handwriting available

## TrOCR + FLAN

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**48.4%**

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## Qwen 2

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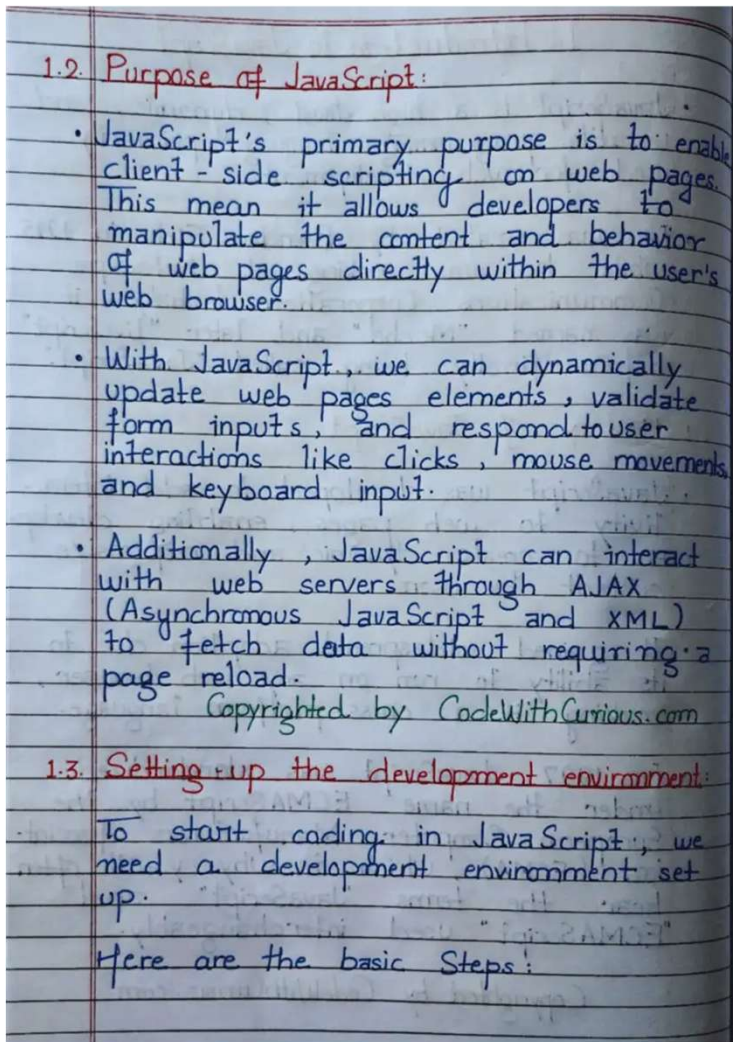
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- Option 1: Hardware for handwriting NOT available

- Option 2: Hardware for handwriting available

**92.8%**

# TEST CASE #2



Qwen 2

1.2. Purpose of JavaScript:

- JavaScript's primary purpose is to enable client-side scripting on web pages. This means it allows developers to manipulate the content and behavior of web pages directly within the user's web browser.
- With JavaScript, we can dynamically update web page elements, validate form inputs, and respond to user interactions like clicks, mouse movements, and keyboard input.
- Additionally, JavaScript can interact with web servers through AJAX (Asynchronous JavaScript and XML) to fetch data without requiring a page reload.

1.3. Setting up the development environment:

To start coding in JavaScript, we need a development environment set up. Here are the basic steps:

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98.8%

# PREScription

Name: Armando Cegua  
 Address: West Rombo, Makati City  
 Age: 29 Sex: M Date: 12-03-90

**Rx**

(Thmox)  
 Amoxicillin Young Cap #21  
 Sig: 1 cap 3x a day for  
 seven days.

Physician's Sig. [Signature]  
 Lic. No. 123457  
 PTR No. 1234567  
 S2 No. \_\_\_\_\_

**Qwen 2**

Name: Armando Cegua

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Age: 29

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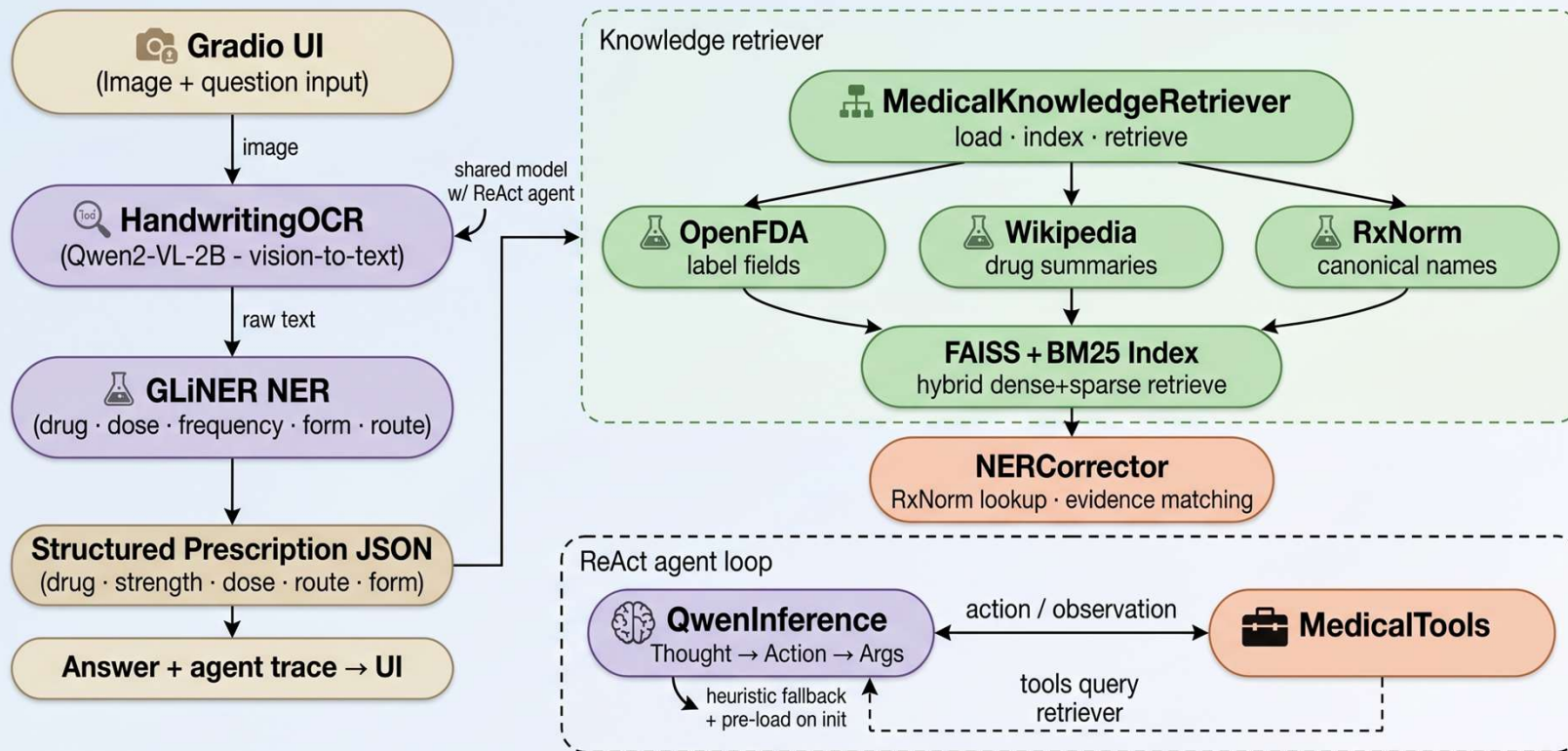
**~80%**





# FULL AGENTIC SYSTEM ARCHITECTURE

## AI Medical Assistant Pipeline: Process and Tool Integration





## TEST CASE

DEA# GB 05455616

LIC # 976269

MEDICAL CENTRE  
824 14<sup>th</sup> Street  
New York, NY 91743, USA

NAME John Smith AGE 34  
ADDRESS 162 Example St, NY DATE 09-11-12

R<sub>x</sub>

Betaloc 100mg - 1 tab BID  
Dorzolamidum 10 mg - 1 tab BID  
Cimetidine 50 mg - 2 tabs TID  
Oxprelol 50mg - 1 tab QD

Dr. Steve Johnson  
signature

☐ LABEL  
REFILL 0 1 2 3 4 5 PRN

WTX-N-PRESC-T  
1-889-422-0700

## EXTRACTED+CORRECTED NER

| Label  | Original    | Corrected                                                                             |
|--------|-------------|---------------------------------------------------------------------------------------|
| drug   | Betaloc     | Betaloc                                                                               |
| drug   | Dorzolamide | Preservative-Free dorzolamide 20 MG/ML / timolol 5 MG/ML Ophthalmic Solution [Cosopt] |
| drug   | Cimetidine  | cimetidine 200 MG Oral Tablet [Tagamet]                                               |
| drug   | Oxperelol   | Oxperelol                                                                             |
| dosage | 50mg        | 50mg                                                                                  |

## USER QUESTION

Ask a Medical Question

What are the side effects of the medicines prescribed?

## AGENT ANSWER

Step 4: Agent Answer

The side effects of the medicines prescribed include nausea, vomiting, diarrhea, and rash.

## **EXPLORING RPA AND AGENTIC RPA**



# RPA

**Robotic Process Automation** mimics human actions on a computer.

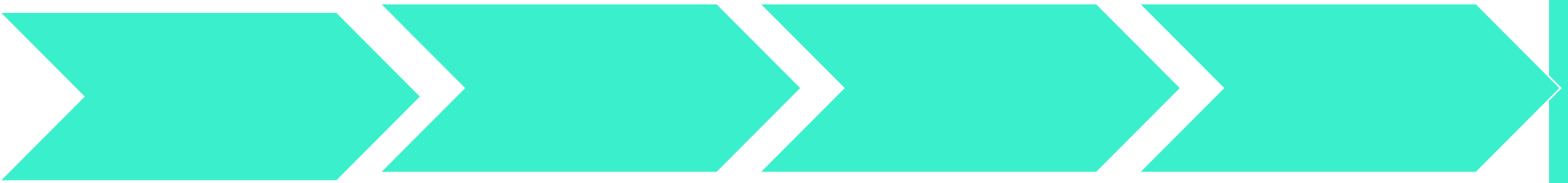
It can:

- Open applications
- Click buttons
- Copy/paste data
- Read emails
- Fill forms
- Download/upload files
- Move data between systems

| Software Tools                                                 | Programmatic Approach                                                                               |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| I. UiPath<br>II. Automation Anywhere<br>III. MS Power Automate | <ul style="list-style-type: none"><li>• Selenium</li><li>• PyAutoGUI</li><li>• Playwright</li></ul> |

**APA (Agentic Process Automation)** adds an AI agent in front of that same automation(For example, *reading and normalization of* the raw data into a clean, structured output

# DOCUMENT PARSING & EXTRACTION TOOLS



# DOCUMENT PARSING & EXTRACTION TOOLS

| Tool       | Function                                                |
|------------|---------------------------------------------------------|
| Mayan EDMS | Electronic Document Management System (EDMS)            |
| Unstract   | Extract structured data from messy documents using LLMs |
| Docling    | Document Parsing                                        |

# SAMPLE TEST CASE

## UNSTRACT

```
{
  "patient": {
    "name": "John Smith",
    "age": 34,
    "address": "162 Example St, NY"
  },
  "date": "2012-09-11",
  "doctor": "Dr. Steve Johnson",
  "prescriptions": [
    {
      "medicine": "Betoloc",
      "dosage": "100mg",
      "instruction": "1 tab BID",
      "frequency": "Twice daily"
    },
    {
      "medicine": "Dorzolamidum",
      "dosage": "10 mg",
      "instruction": "1 tab BID",
      "frequency": "Twice daily"
    }
  ]
}
```

## DOCLING

# Medical Centre

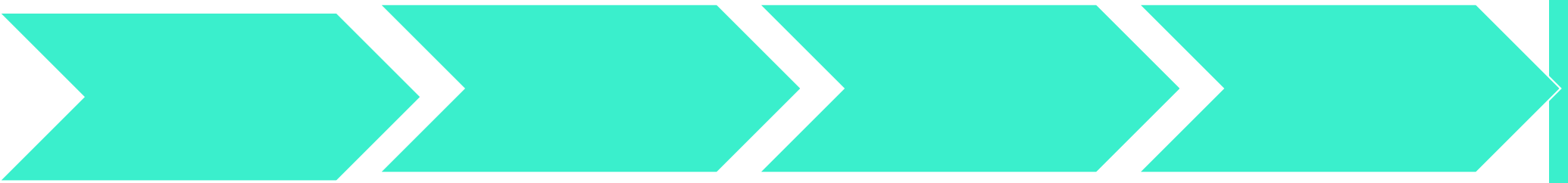
## Patient Information

- Name: John Smith
- Age: 34
- Address: 162 Example St, NY

## Prescription

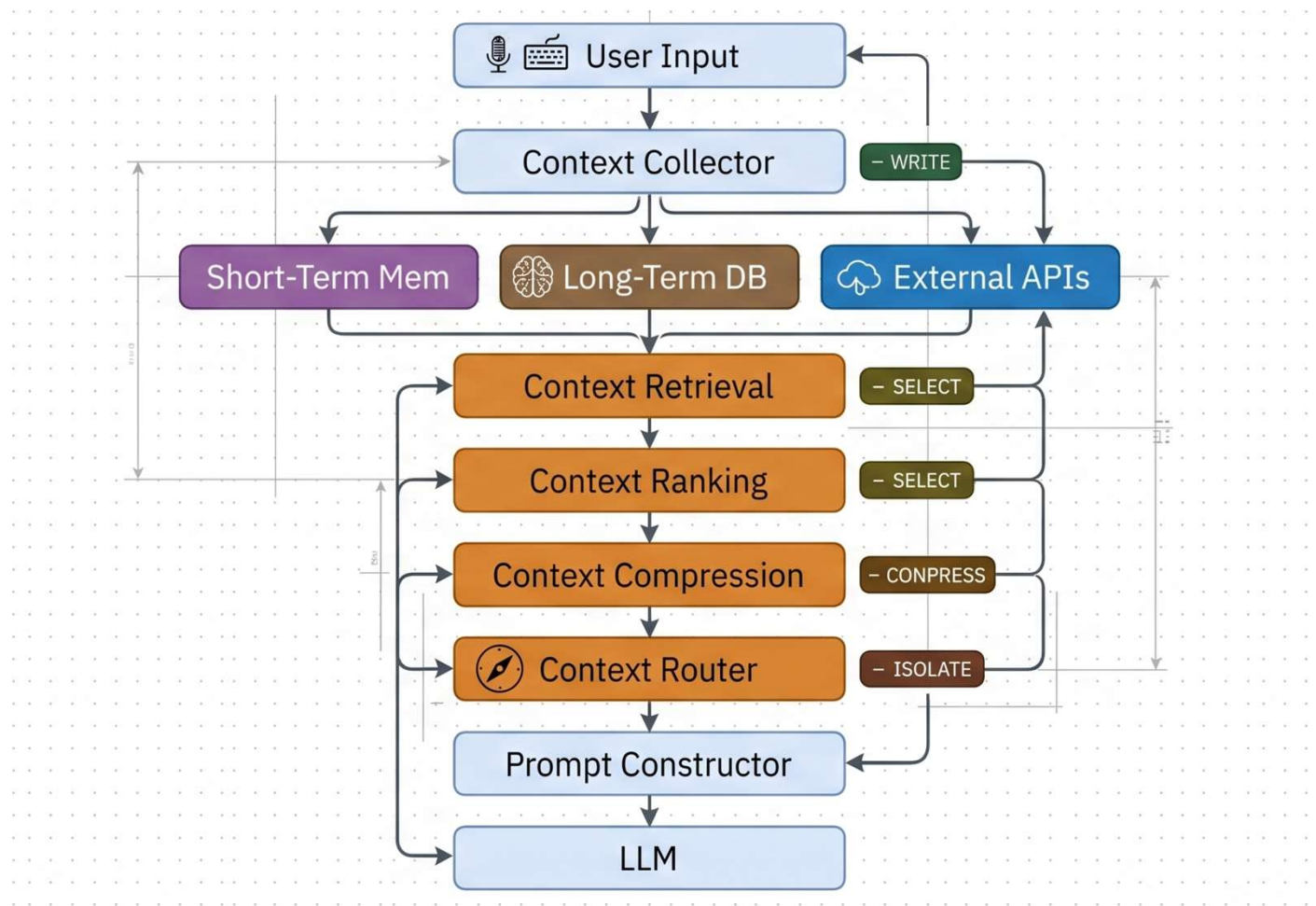
- Betoloc 100mg - 1 tab BID
- Dorzolamidum 10 mg - 1 tab BID

# CONTEXT ENGINEERING FRAMEWORK

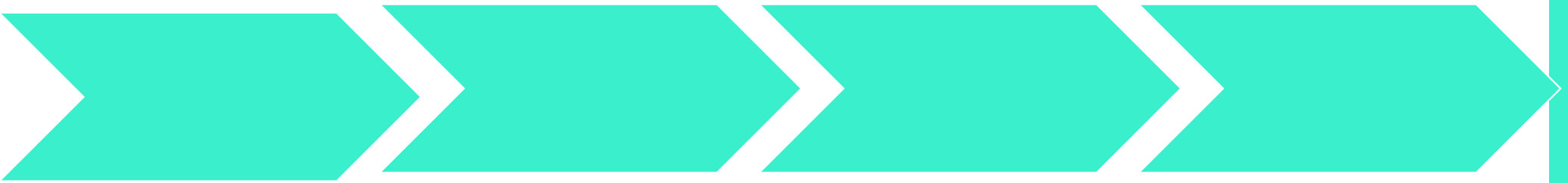




# CONTEXT ENGINEERING FRAMEWORK

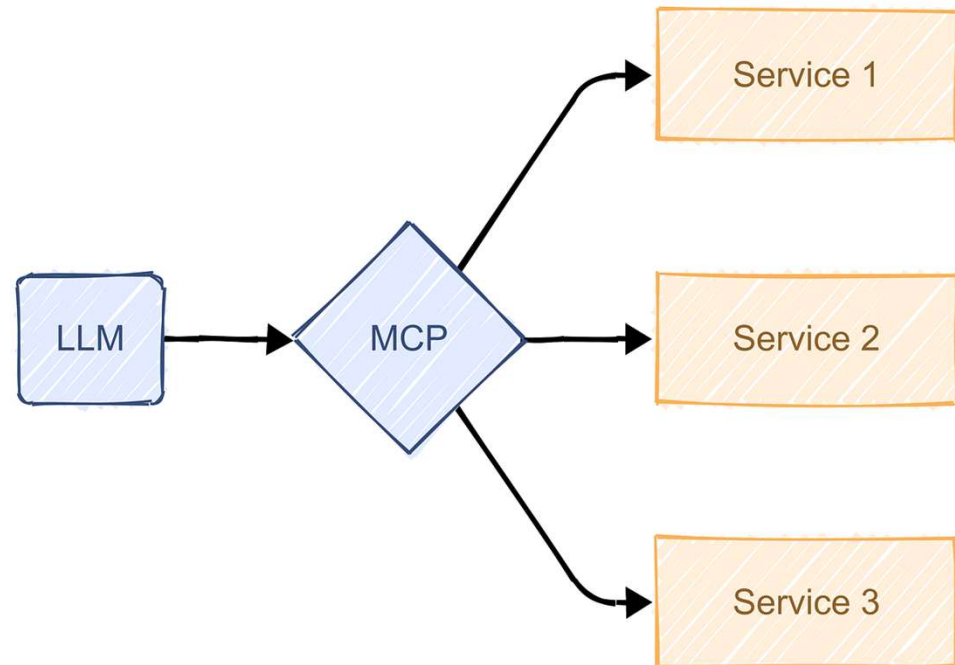


# IMPLEMENTING MODEL CONTEXT PROTOCOL



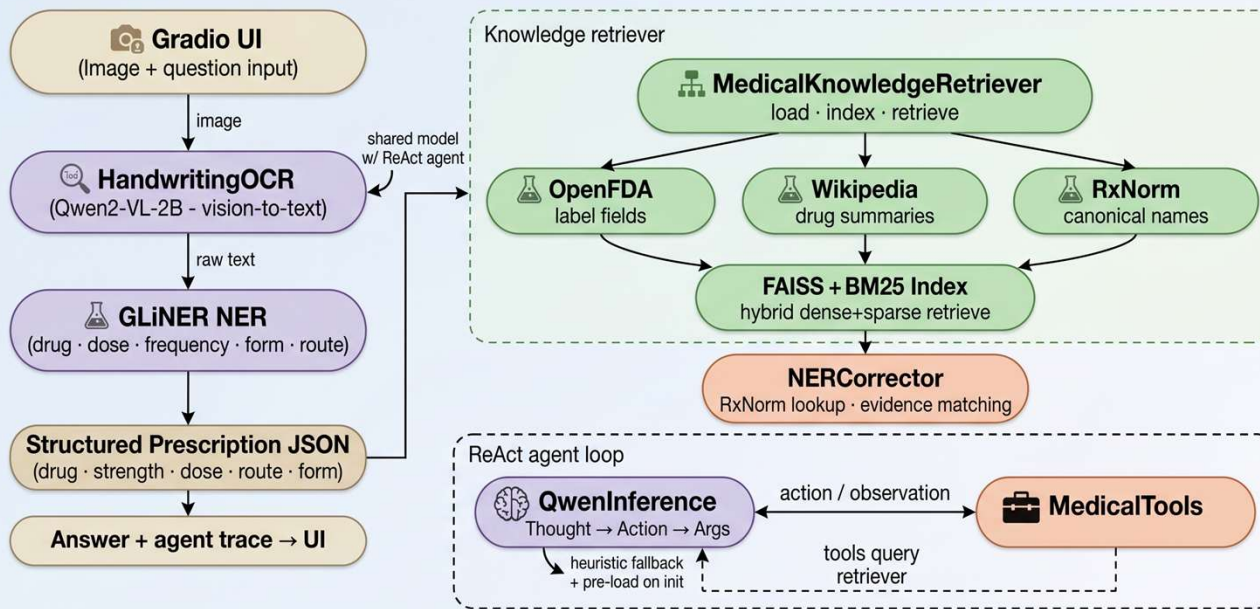
# M C P

- **Agents** can't DO things without function tools/APIs.
- Every tool needs a custom integration (no standard).
  - Model Context Protocol(MCP)
- One open protocol to expose Tools, Resources & Prompts



# ARCHITECTURE W/O MCP

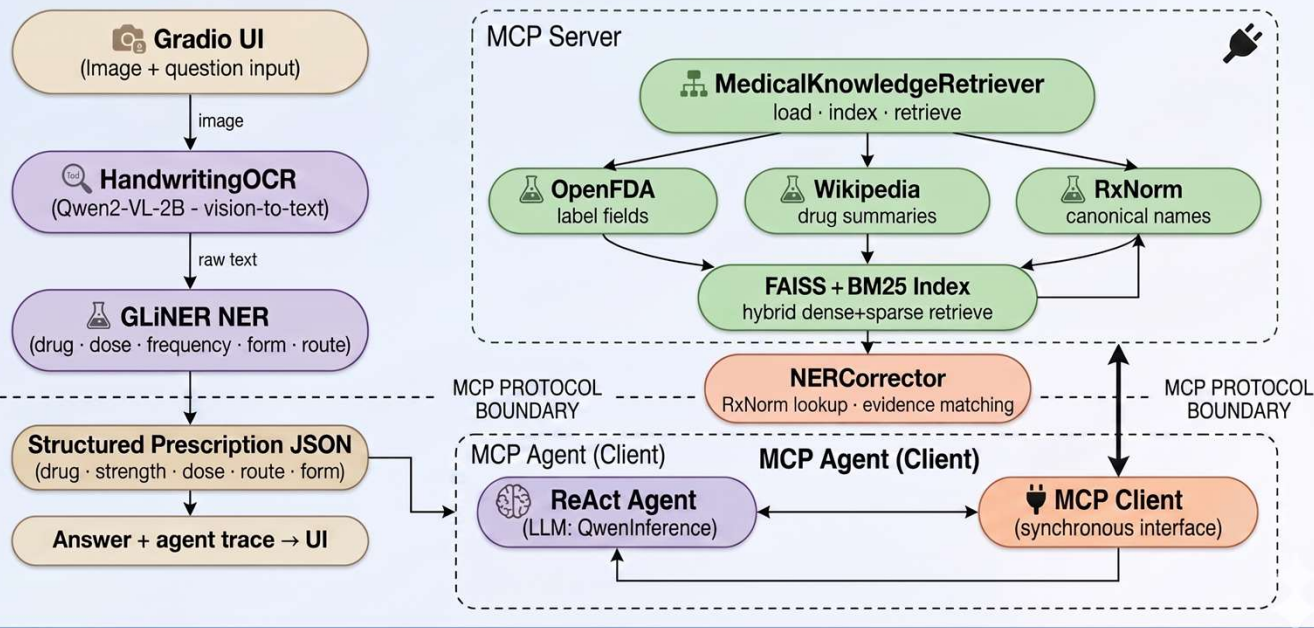
## AI Medical Assistant Pipeline: Process and Tool Integration



- Tightly connected
- Agent owns retrieval logic
- Not scalable
- No multi-agent support

# WITH MCP

## AI Medical Assistant Pipeline: Process and Tool Integration



- Agent doesn't care about source of retrieval
- Agent owns only tool usage
- Other agents can also use it
- Scalable
- Multi-agent support

## ORIGINAL — Direct Python Dispatch

### ① Constructor — agent owns tools directly

```
class ReActAgent:
    def __init__(self, qwen,
                  tools: MedicalTools):
        self.tools = tools # object ref
```

### ② Tool invocation — plain Python method

```
# Inside run() loop:
obs = self.tools.dispatch(
    action, args)
```

### ③ Hard-coded tool registry — no schemas

```
tool_map = {
    "search_knowledge_base":
        lambda a: self.search_kb(a),
    "fetch_openfda_field":
        lambda a: self.fetch_fda(a),
}
```

### ④ No tool discovery — names hard-coded

```
# Agent must know tool names
# at development time. No runtime discovery.
```

## MCP — Protocol-Based Tool Call

### ① Constructor — agent holds MCPClient

```
class ReActAgent:
    def __init__(self, qwen,
                  mcp_client: MCPClient):
        self.mcp_client = mcp_client
```

### ② Tool invocation — MCP protocol message

```
# Inside run() loop:
obs = self.mcp_client.call_tool_sync(
    action, args)
```

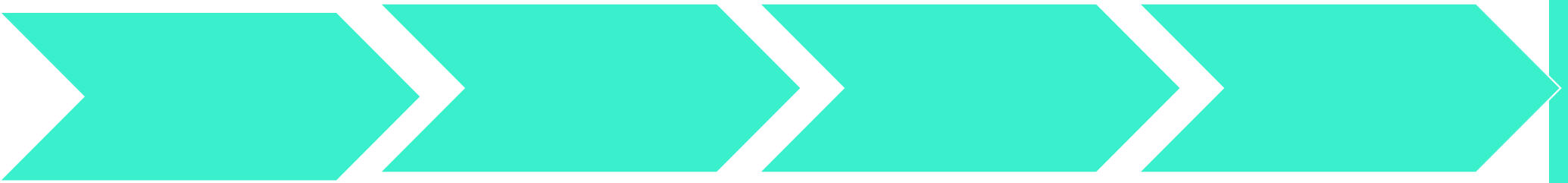
### ③ Schema-declared tool registry

```
@mcp_server.list_tools()
async def list_tools():
    return TOOL_SCHEMAS # JSON schemas
@mcp_server.call_tool()
async def call_tool(name, args):
    ...
```

### ④ Runtime tool discovery via protocol

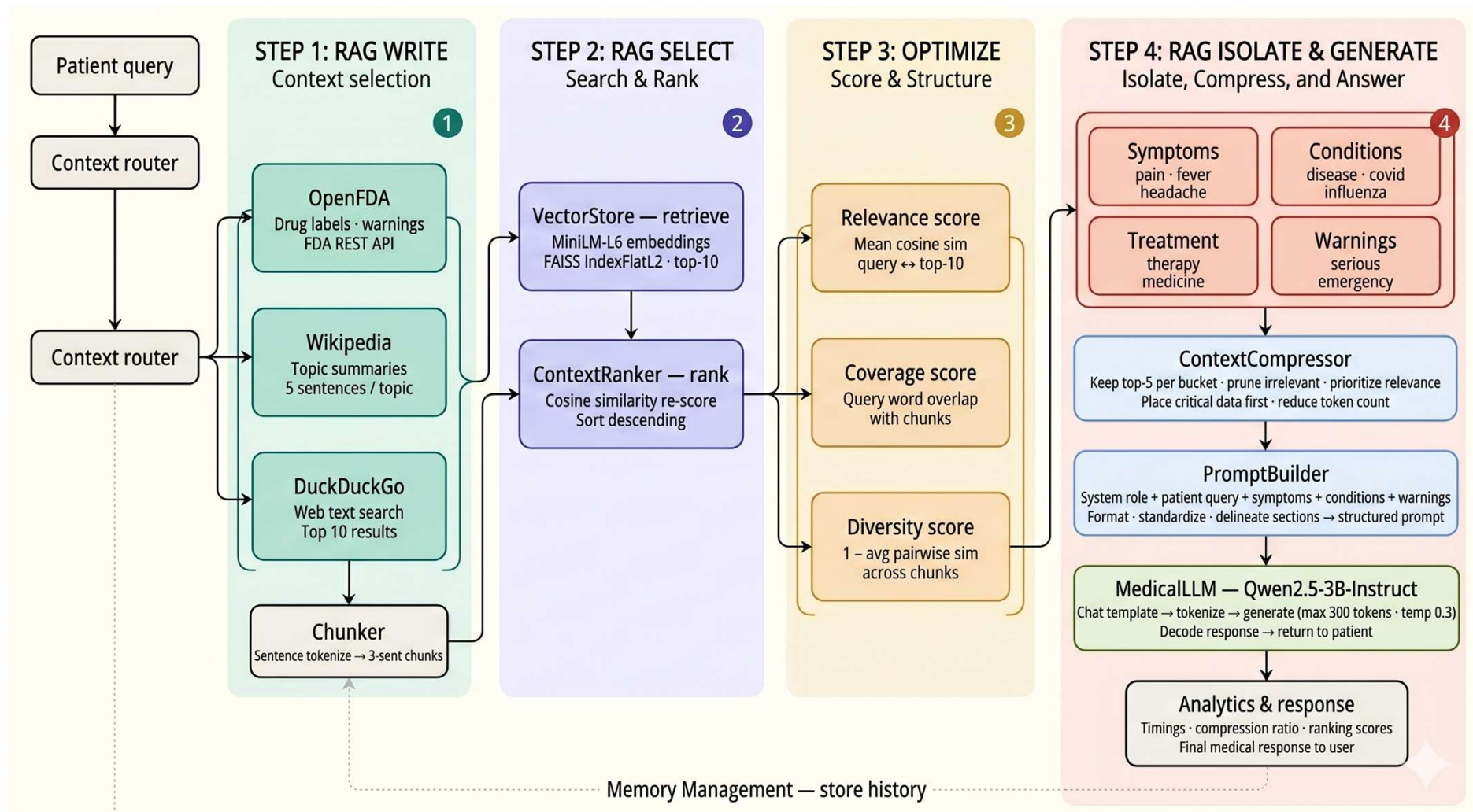
```
tools = await client.list_tools()
# Agent discovers tools at runtime.
```

# **BUILDING CONTEXT ENGINEERING FRAMEWORK**





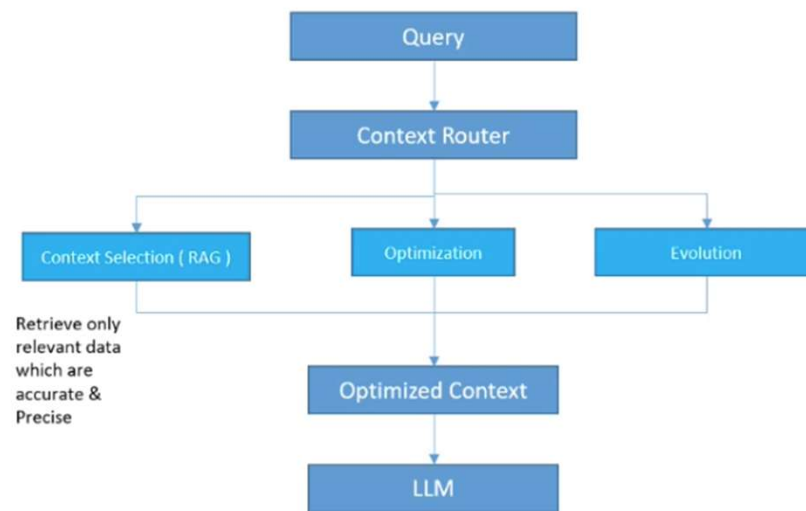
# CONTEXT ENGINEERING FRAMEWORK





# CONTEXT ENGINEERING FRAMEWORK

## Context Engineering



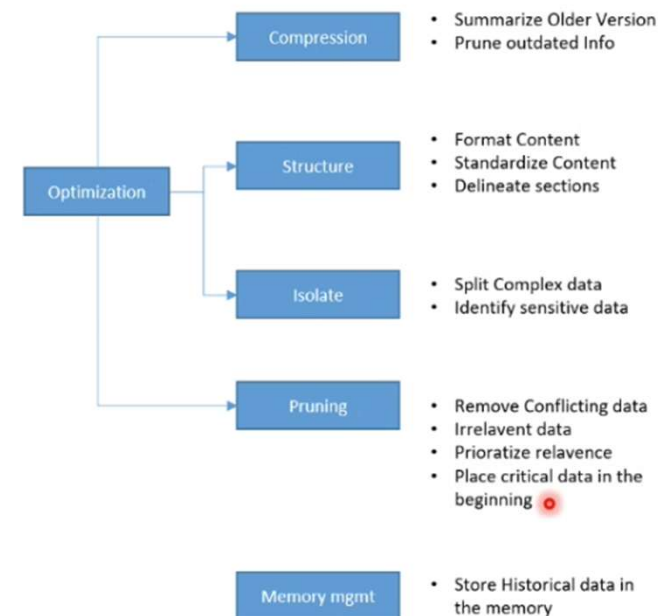
### Harness Engineering Vs Context Engg

<https://www.youtube.com/watch?v=1a1VXDdlyrk>

[https://www.youtube.com/watch?v=C\\_GG5g38vLU](https://www.youtube.com/watch?v=C_GG5g38vLU)

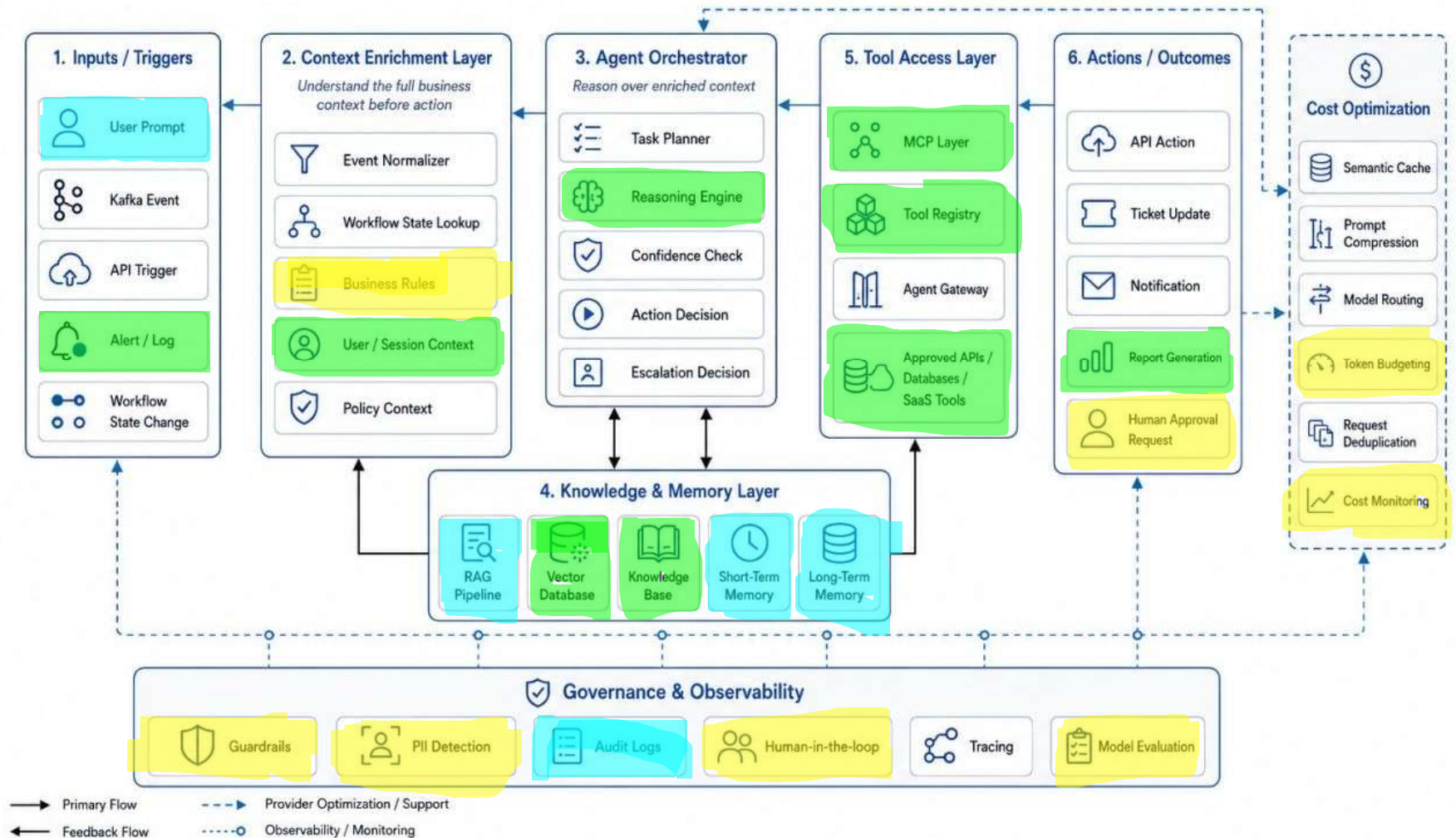
<https://www.langchain.com/blog/the-anatomy-of-an-agent-harness>

<https://www.linkedin.com/pulse/context-engineering-vs-harness-next-evolution-ai-systems-imdad-areeph-h4c2f/>



# Context Engineering Architecture for Agentic AI Systems

Reliable enterprise agents with context enrichment, governance, and cost optimization



#### Patient Query

I'm having coughs and headaches for the past 4 days. What could be causing it?

### WITH RAW CONTEXT

#### Textbox

Based on your symptoms of coughing and headaches over the past four days, there are several potential causes that could be contributing to your discomfort. However, it's important to note that without a proper examination by a healthcare professional, it's difficult to provide a definitive diagnosis. Here are some possibilities:

1. **"Common Cold or Influenza"**: Both can cause coughing and headaches. The common cold is usually caused by viruses, while influenza (flu) is also viral but tends to cause more severe symptoms.
2. **"Allergies"**: Allergic reactions can cause both coughing and headaches. Common allergens include pollen, dust mites, pet dander, and mold.
3. **"Bronchitis"**: This is inflammation of the airways in the lungs and can cause coughing and headaches, especially if it's a result of a viral infection like the flu.
4. **"Sinusitis"**: Inflammation of the sinuses can lead to coughing and headaches, particularly if it's due to a bacterial infection.
5. **"Gastroesophageal Reflux Disease (GERD)"**: Acid reflux can irritate the throat and airways, leading to coughing and headaches.
6. **"Dehydration"**: Not drinking enough water can lead to dehydration, which might manifest as headaches and coughing.

Given the warnings and precautions mentioned [in the product information](#), it seems you should avoid alcohol and other substances that can affect your nervous system or cause drowsiness while taking

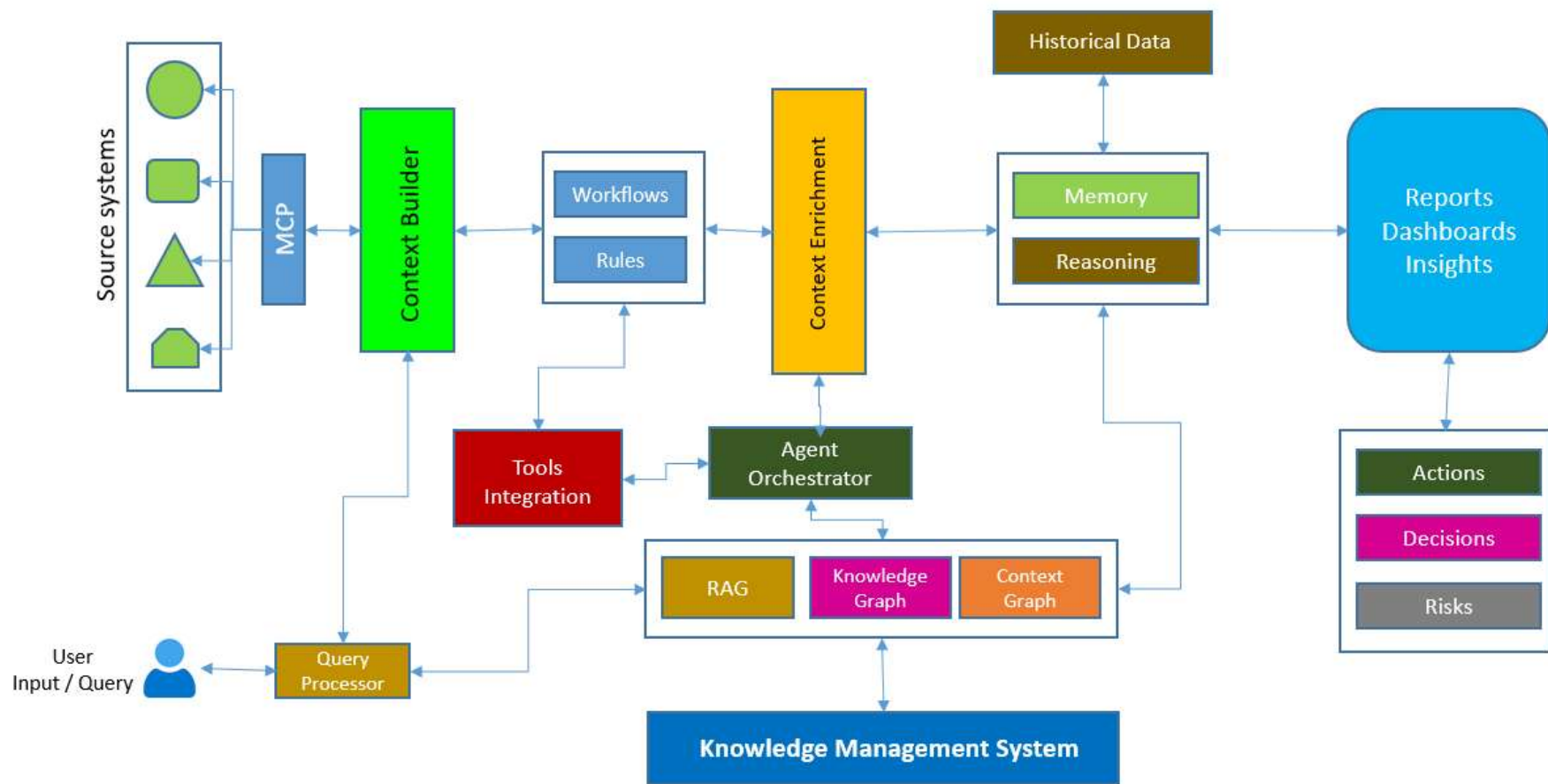
### WITH CONTEXT ENGINEERING

#### Textbox

Based on the symptoms of coughing and headaches lasting four days, there are several potential causes. These could include viral infections like the common cold or flu, allergies, or even less common conditions like sinusitis or bronchitis. The severity can vary; some cases might resolve on their own, while others may require medical attention depending on how they progress.

Warning signs that should prompt a visit to a healthcare provider include persistent or worsening symptoms, fever, difficulty breathing, severe pain, or changes in vision. Additionally, if you experience severe dizziness, sleeplessness, or nervousness, these could indicate a more serious issue.

For next steps, I recommend consulting with a healthcare professional. They can perform a physical examination and possibly order tests to determine the exact cause of your symptoms. In the meantime, stay hydrated, rest, and avoid irritants that might exacerbate your symptoms. If your symptoms worsen or persist, seek medical advice promptly.



Pipeline (runs top to bottom)

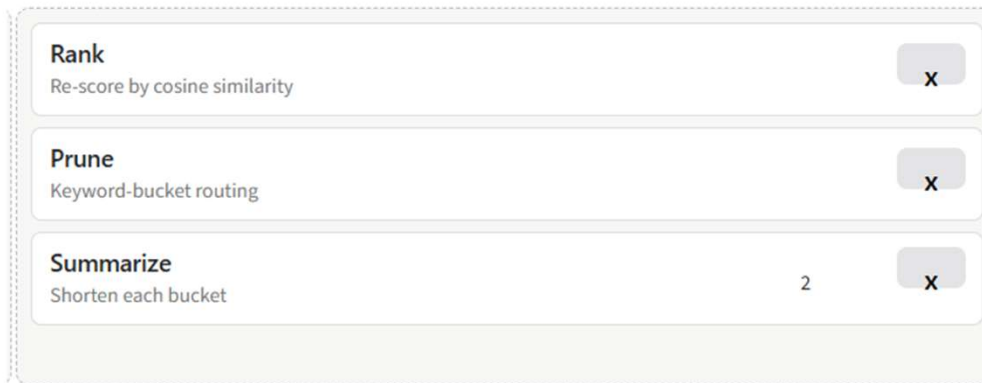


Textbox

1. Likely causes of constant headaches could include migraines, tension-type headaches, or even an underlying viral infection like the flu. Since there are no red flags, we need to consider the symptoms and potential triggers.
2. The severity can vary widely. Migraines can be debilitating, while occasional tension headaches might be more manageable. Viral infections like the flu can cause severe headaches along with other flu-like symptoms.
3. Warning signs to watch out for include sudden, severe headaches, difficulty breathing, or any changes in vision. These could indicate a more serious condition such as a brain tumor or stroke.
4. Next steps would involve discussing your symptoms with a healthcare provider. They might recommend over-the-counter pain relievers for mild cases or prescribe medications like gabapentin if needed. If your headaches are severe or accompanied by other concerning symptoms, further diagnostic tests might be necessary. Always follow up with a healthcare professional for proper evaluation and treatment.



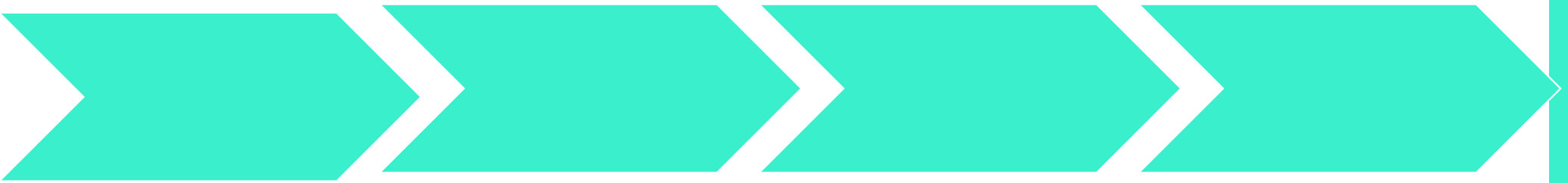
Pipeline (runs top to bottom)



Textbox

1. Likely causes of constant headaches could include migraines, tension-type headaches, or even an underlying viral infection like the flu. Migraines are often associated with warning signs such as visual disturbances or nausea. The flu, while less common, can cause headaches as a symptom.
2. The severity of headaches can vary widely from mild to severe. For instance, migraines can be debilitating, causing significant pain and affecting daily activities. On the other hand, occasional tension-type headaches might be more tolerable but still disruptive.
3. Warning signs to watch out for include sudden, severe headaches, which could indicate a more serious condition like a brain tumor or aneurysm. Other signs include difficulty breathing, which is a serious warning sign that requires immediate medical attention.
4. Next steps would involve discussing your symptoms with a healthcare provider. They may recommend over-the-counter pain relievers for mild headaches or prescribe medications like gabapentin if necessary. If the headaches are severe, persistent, or accompanied by other concerning symptoms, further diagnostic tests or referrals to specialists might be needed. It's important to monitor your symptoms and seek medical advice promptly if they worsen or change significantly.

# **PATIENT HANDBOOK SYSTEM IMPLEMENTATION**

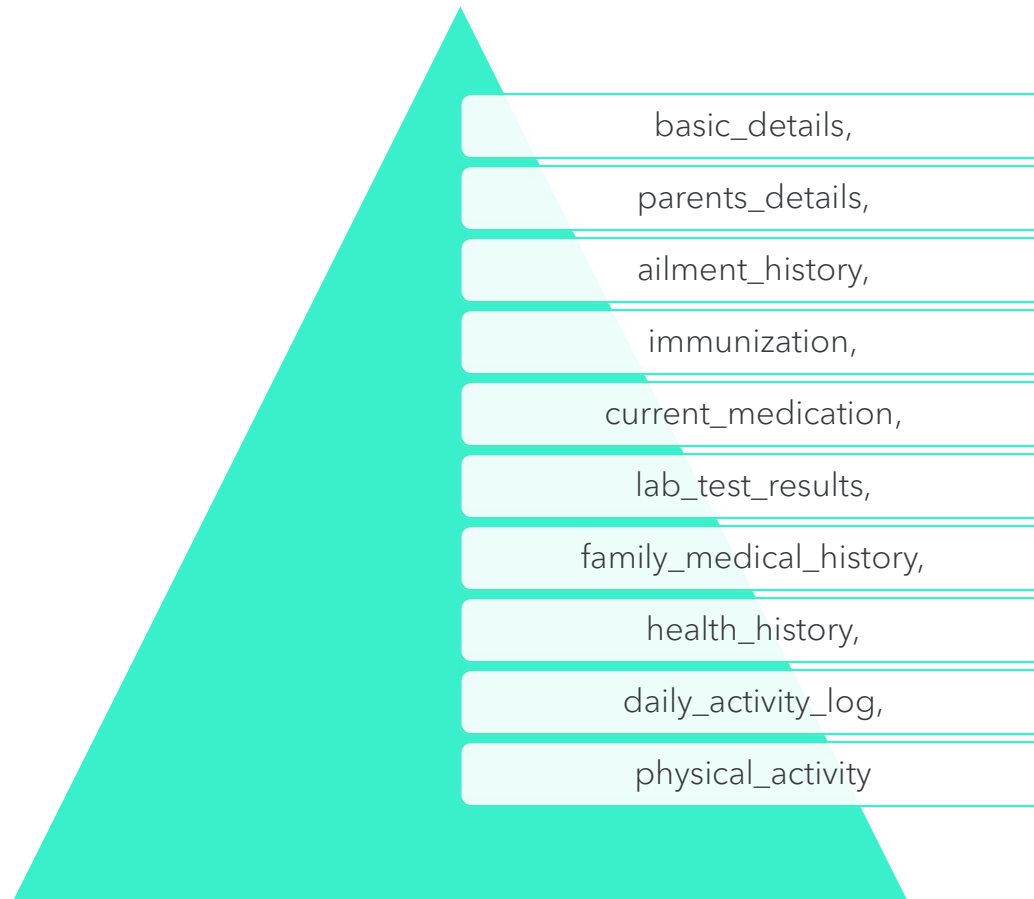


# PATIENT HANDBOOK

| Source Layer         | Dataset                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Patient Record / EHR | Synthetic Data( <i>Faker</i> )                                                                                                       |
| Lab Reports          | Synthetic Data                                                                                                                       |
| Historical Reports   | Synthetic Data                                                                                                                       |
| Medical Images       | NIH Chest X-Ray Dataset<br><a href="https://kaggle.com/datasets/nih-chest-xrays">kaggle.com/datasets/nih-chest-xrays</a>             |
| Voice Conversation   | MedQuad<br><a href="https://keivalya/MedQuad-MedicalQnADataset">keivalya/MedQuad-MedicalQnADataset</a>                               |
| Email / Chat         | MedDialog Dataset<br><a href="https://github.com/UCSD-AI4H/Medical-Dialogue-System">github.com/UCSD-AI4H/Medical-Dialogue-System</a> |
| EHIR Data            | <i>Synthesized based on patient record</i>                                                                                           |
| Smart Watch          | Synthetic Data                                                                                                                       |
| Web / Public         | MedQA / PubMedQA<br><a href="https://kaggle.com/datasets/pythonaio/pubmedqa">kaggle.com/datasets/pythonaio/pubmedqa</a>              |



# PATIENT HANDBOOK DESIGN



# XML BUILDER

```
<PATIENT_PASSBOOK>

<BASIC_DETAILS>
{json.dumps(patient["basic_details"], indent=2)}
</BASIC_DETAILS>

<PARENTS_DETAILS>
{json.dumps(patient["parents_details"], indent=2)}
</PARENTS_DETAILS>

<AILMENT_HISTORY>
{json.dumps(patient["ailment_history"], indent=2)}
</AILMENT_HISTORY>

<IMMUNIZATION>
{json.dumps(patient["immunization"], indent=2)}
</IMMUNIZATION>

<CURRENT_MEDICATION>
{json.dumps(patient["current_medication"], indent=2)}
</CURRENT_MEDICATION>

<LAB_TEST_RESULTS>
{json.dumps(patient["lab_test_results"], indent=2)}
</LAB_TEST_RESULTS>

<FAMILY_MEDICAL_HISTORY>
{json.dumps(patient["family_medical_history"], indent=2)}
</FAMILY_MEDICAL_HISTORY>

<HEALTH_HISTORY>
{json.dumps(patient["health_history"], indent=2)}
</HEALTH_HISTORY>

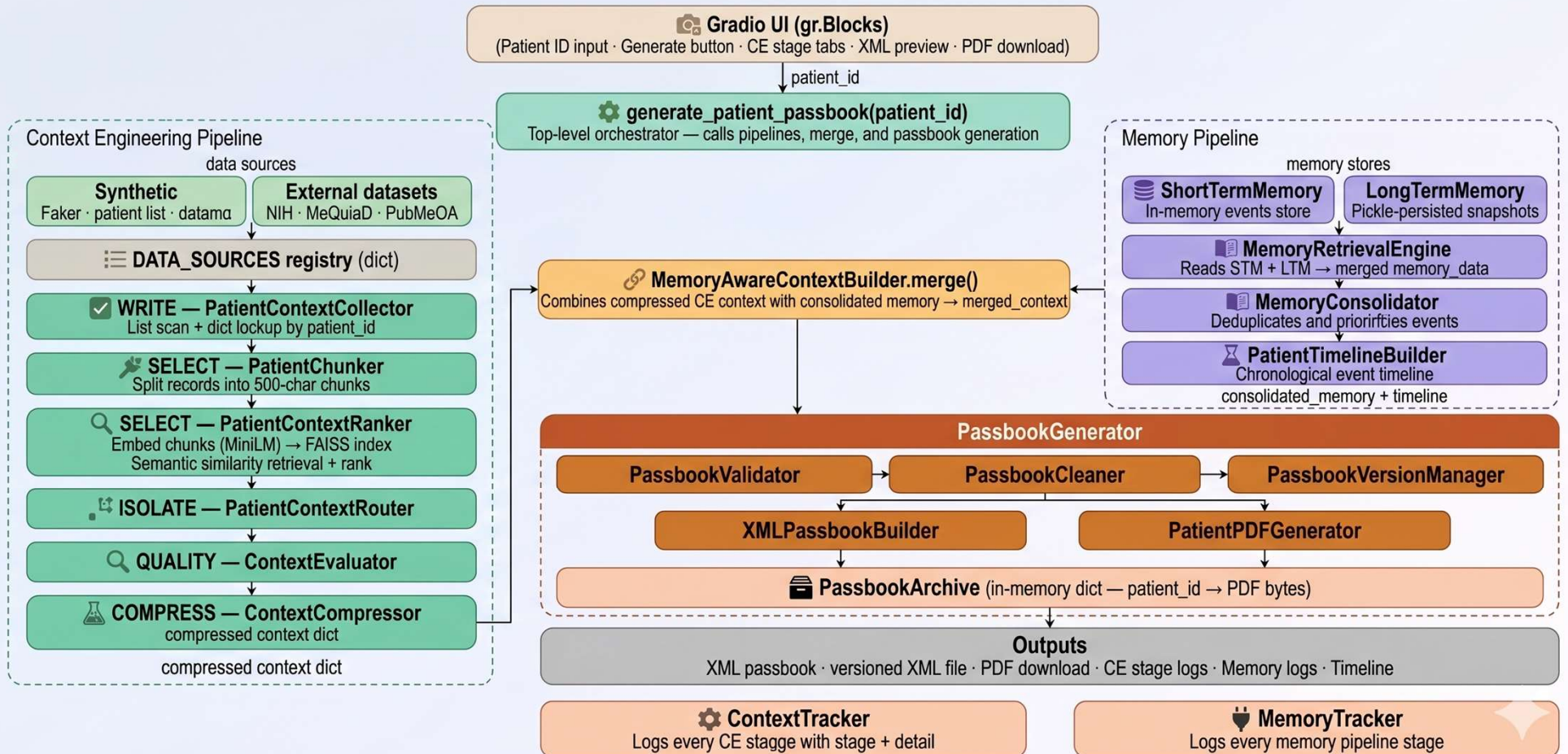
<DAILY_ACTIVITY_LOG>
{json.dumps(patient["daily_activity_log"], indent=2)}
</DAILY_ACTIVITY_LOG>

<PHYSICAL_ACTIVITY>
{json.dumps(patient["physical_activity"], indent=2)}
</PHYSICAL_ACTIVITY>

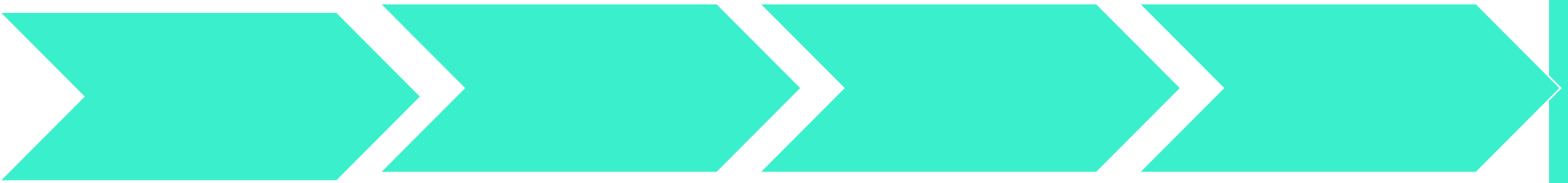
</PATIENT_PASSBOOK>
```

# PATIENT HANDBOOK

## Context-Aware Patient Passbook Pipeline: Stage-by-Stage Architecture and Components

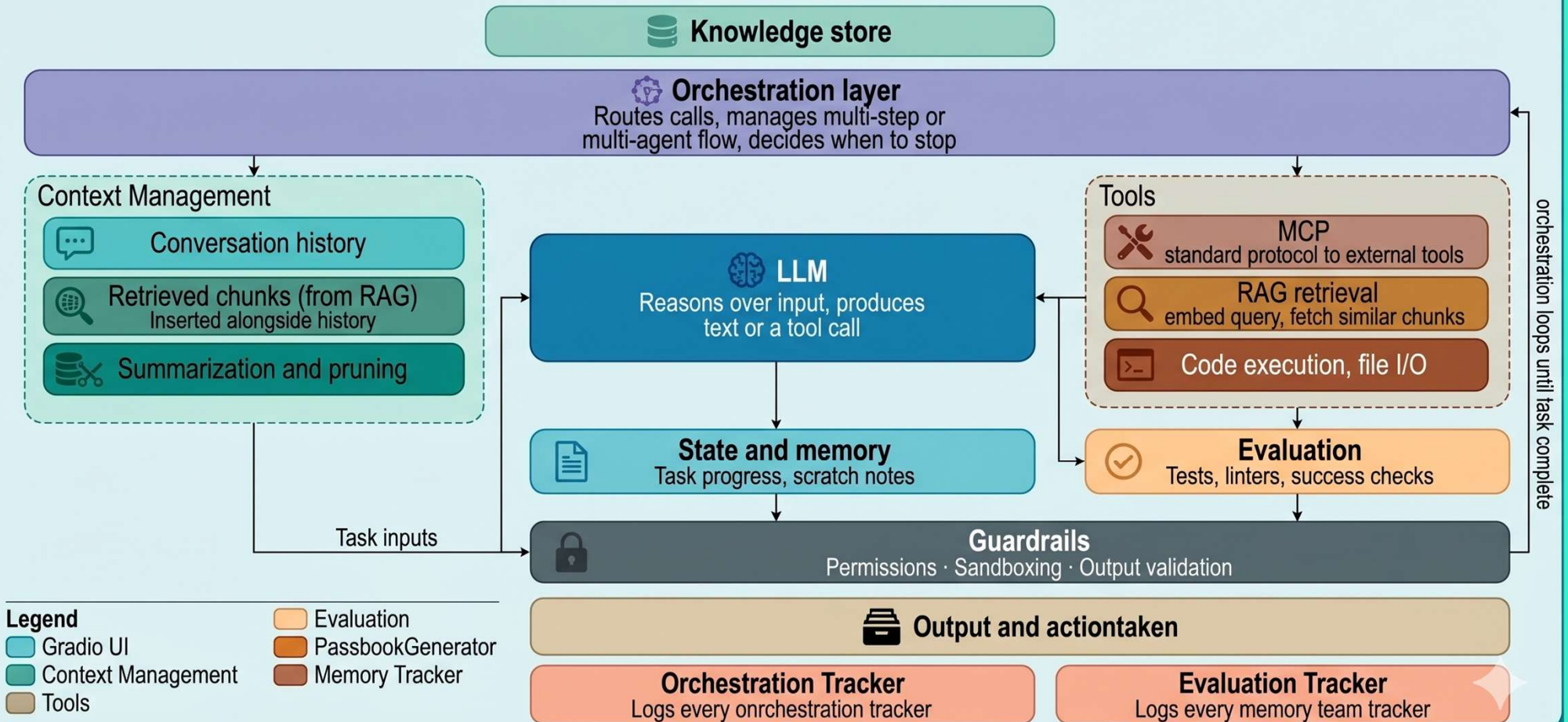


## **EXPLORING HARNESS ENGINEERING**



# AI Harness System Architecture for Stage-by-Stage Processing and Control

Everything engineered around the model



Version: 1.2.3 | © 2022 Markotopies, Inc.



**Harness engineering is the outer layer of design, including prompt and context engineering.**

Outer layers encompass broader scope and design.

## **Harness Engineering**

Designing the complete working structure, including objectives, procedures, records, reviews, and roles.

## **Context Engineering**

Designing how to provide the necessary information to the AI.

## **Prompt Engineering**

Crafting specific, individual instructions for the AI.

Outer layers encompass broader scope and design.